

## 6.8 Steps 10 – 12: Data Analysis, Conclusion and Recommendations

دسویں قدم میں چارٹس اور اسپریڈشیٹ سے رجحانات تلاش کیے جاتے ہیں: زیادہ تر طلبہ روزانہ تین سے پانچ گھنٹے فون است...

CS XI

Chapter 6: Digital Literacy

Lecture 15 of 17

★ Golden

### 6.8 Steps 10 – 12: Data Analysis, Conclusion and Recommendations

★ Golden

#### Step 10: Analyse the results

##### LEARN IT

To **analyse** means to study the data carefully to find **patterns** and **trends**. By looking at the spreadsheet and charts, the group identifies:

- **Most students** (the majority) use phones for **3–5 hours daily**.
- **Half** use it for **educational purposes** (lectures/notes).
- **A clear trend:** students who use phones mostly for **entertainment** report much **higher distraction** levels than those who use them for study.

#### Step 11: Write the conclusion

##### LEARN IT

A **conclusion** answers the research question using the evidence:

*"The research shows mobile phones have both positive and negative effects. They provide access to learning materials. However, excessive use of social media and gaming reduces concentration. Therefore, phones should not be banned, but managed wisely."*

#### Step 12: Recommendations

##### LEARN IT

**Recommendations** are practical suggestions based on the data. The students recommend:

- Turn off **social media notifications** while studying.
- Set **daily screen-time limits**.
- Take **physical breaks** from screens to protect health.

##### WORKED EXAMPLE

**From numbers to a finding.** Using the estimated chart counts (12, 21, 7 out of 40):

**Step 1:**  $21 \div 40 = 52.5\%$ . Since  $52.5\% > 50\%$ , the statement "most students use phones for 3–5 hours" is supported.

**Step 2:** "Half use it for educational purposes" means about 20 of the 40 students chose Lectures or Notes as their main activity ( $20 \div 40 = 50\%$ ). Formula: `=COUNTIF(D2:D41, "Lectures")+COUNTIF(D2:D41, "Notes")`.

**Step 3 – Compare groups:** filter the sheet to show only Social Media/Games users and count how many said "Yes" to distracted; then do the same for Lectures/Notes users. If the entertainment group has a much higher share of "Yes", that is the trend.

**Step 4 – Link to secondary data:** the article said excessive screen time lowers concentration; the survey agrees, so the conclusion is stronger.

#### REMEMBER IT

Conclusion formula: "**The data shows ... However ... Therefore ...**" – state the main finding, mention the other side, then give the overall answer.

#### COMMON MISTAKE

**Mistake:** Writing a conclusion that is not based on the data, e.g. "Phones should be banned" when the data showed phones also help learning. **Fix:** Every sentence in the conclusion and recommendations must be supported by your findings. Also, a trend between two things (entertainment use and distraction) shows a *link*; be careful before claiming one definitely *causes* the other.

#### اردو میں سمجھیں

دسویں قدم میں چارٹس اور اسپریڈشیٹ سے رجحانات تلاش کیے جاتے ہیں: زیادہ تر طلبہ روزانہ تین سے پانچ گھنٹے فون استعمال کرتے ہیں، نصف اسے تعلیم کے لیے استعمال کرتے ہیں، اور تفریح کے لیے استعمال کرنے والے زیادہ توجہ بٹنے کی شکایت کرتے ہیں۔ گیارہویں قدم میں نتیجہ لکھا جاتا ہے کہ فون پر پابندی نہیں بلکہ اس کا دانشمندانہ استعمال ہونا چاہیے۔ بارہویں قدم میں سفارشات دی جاتی ہیں: پڑھائی کے دوران نوٹیفکیشن بند کریں، اسکرین ٹائم کی حد مقرر کریں اور وقفے لیں۔

#### EXAM TIP

You may be asked to "write a conclusion and two recommendations" for given data. Use the "shows ... however ... therefore" pattern and make sure each recommendation follows from a finding.

#### CHECK YOURSELF

1. According to Step 10, which group reported higher distraction?
2. Should phones be banned according to the conclusion?
3. Give two of the students' recommendations.

Answers: 1. Students who use phones mostly for entertainment. 2. No – they should not be banned but managed wisely. 3. Any two: turn off social media notifications while studying; set daily screen-time limits; take physical breaks from screens.

Next lecture: 6.8 The Final Digital Artefact